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STRUCTURES AND PROPERTIES OF COMPLEX LIPIDS

I-INTRODUCTION

-Complex lipids: these are heterolipids that contain phosphorus (P), nitrogen (N), or sulfur (S) in addition to C, H, and O.

-They are formed from an alcohol that binds to a fatty acid and/or other compounds.

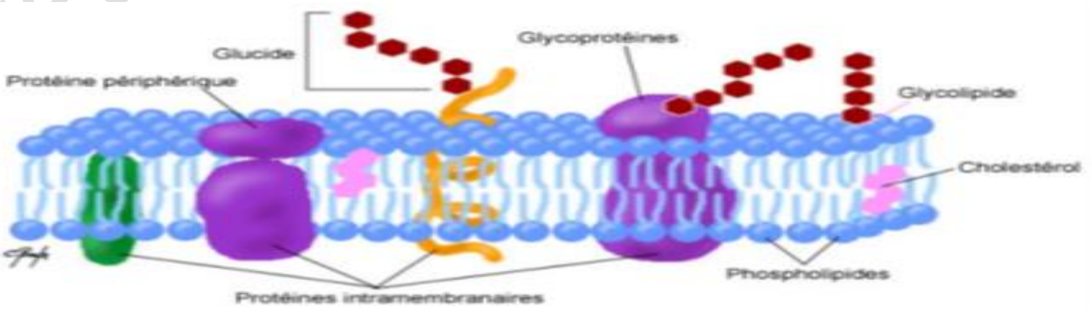
-They can be classified according to the alcohol:

- Either **glycerol**: we distinguish between: Glycerophospholipids (phosphorus)

  Glyceroglycerolipids (sugar)
- Or an **amino alcohol**: Sphingosine, which defines sphingolipids.

-Complex lipids are **essential components** of **biological membranes.**

-However, they have **no energetic role**.



II/GLYCEROPHOSPHOLIPIDS (or PHOSPHOGLYCERIDES)

-**Phospholipids** or **glycerophospholipids** are lipid compounds containing phosphorus.

-They are the most numerous representatives of complex lipids.

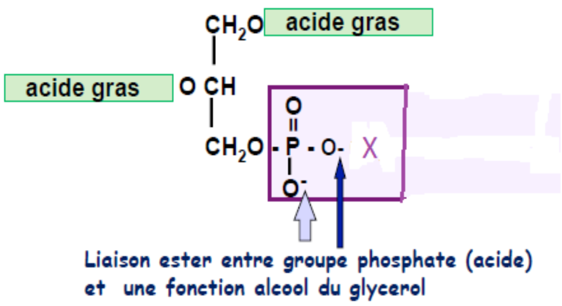
-They are found in high concentrations in biological membranes.

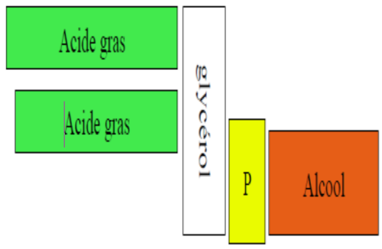
A/Structure:

-They are glycerol esters containing: 1 phosphate, 2 fatty acids, 1 alcohol x

-Phospholipids = 2 fatty acids + glycerol + phosphate + alcohol X

-The general formula is written as follows





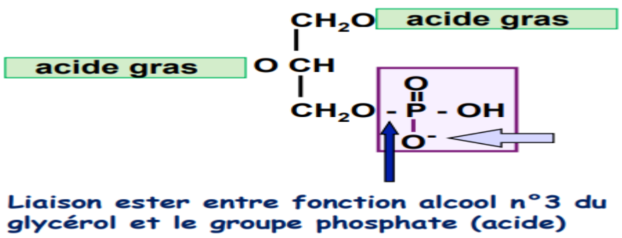
1)Phosphatidic acids:

-These are **phosphoric esters of diglyceride**, meaning that glycerol is esterified by two fatty acids and phosphoric acid.

-In general, fatty acids have between 16 and 18 carbon atoms. Often, one of the fatty acids is saturated and located at position C1, while the other is unsaturated and located at position C2.

-Overall negative charge (acidic phospholipids)

They rarely exist in a **free state** and **play an important** role in the **biosynthesis** of glycerophospholipids.



-Depending on the alcohol that esterifies the phosphatidic acid, different classes are obtained:

2) Nitrogenous glycerophospholipids:

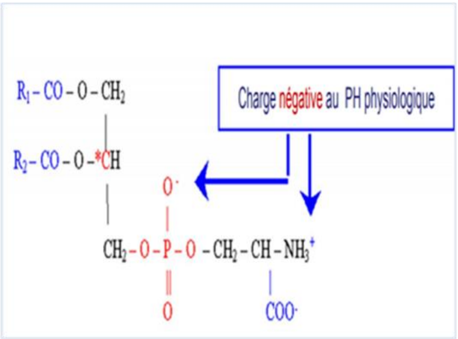
-**Phosphatidylserines** = Phosphatidic acids + **Serine**

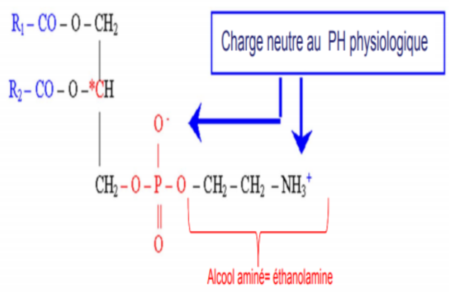
-**Phosphatidylethanolamines** = Phosphatidic acids + **Ethanolamine** (cephalins)

-**Phosphatidylcholines** = Phosphatidic acids + **Choline** (lecithins)

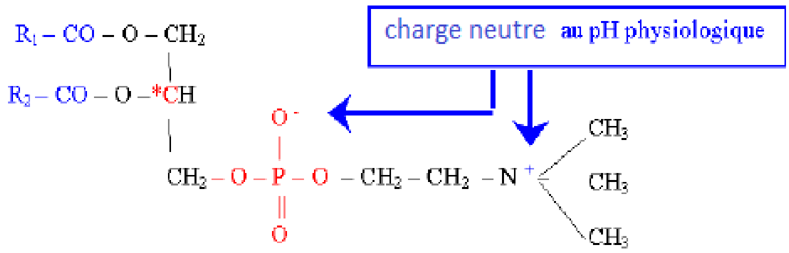
- **Cephalins** are present in all animal tissues and many plant tissues. They are abundant in the brain (hence their name).
- **Lecithins** are one of the constituents of **egg yolk** (oleic acid in 2, stearic acid in 3), but also of liver, kidney, muscle, and nerve tissue cells.

Phosphatidylsérine **Phosphatidyléthanolamine**





Phosphatidylcholine



3) Non-nitrogenous glycerophospholipids:

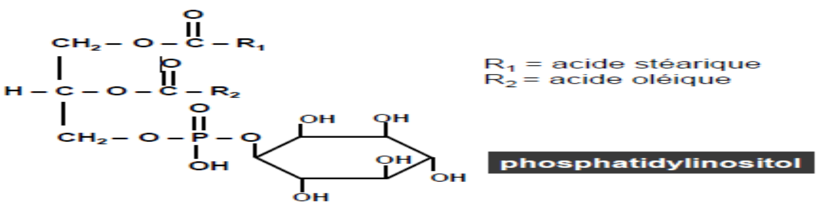
- Phosphatidylglycerols = Phosphatidic acids + **glycerol**

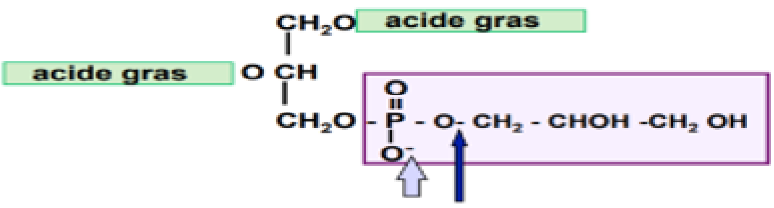
They are abundant in certain microorganisms and plants.

Overall negative charge (acidic phospholipids)

-Phosphatidylinositols (inositides) = Phosphatidic acids + **inositol**

Phosphatidylinositol is located on the inner surface of plasma membranes, where it is the metabolic precursor of two secondary messengers: diglycerides and inositol triphosphate.





Ester bond between phosphate (acid) group and an alcohol function of glycerol

Phosphatidylglycérol

B/PHYSICAL PROPERTIES

1-Polarity and amphiphilic nature:

Glycerophospholipids have **a polar head** (PO4—X) and **a nonpolar tail** (acyl group): they are **amphipathic** molecules

They organize themselves to form **micelles** where the polar heads are in contact with the aqueous medium while the nonpolar tails are directed towards the hydrophobic medium.

2-Amphoteric molecules: an **acidic function** (provided by H3PO4)
a **basic function** (provided by alcohol)

C/CHEMICAL PROPERTIES

1/Chemical hydrolysis:

- **Acid hydrolysis** : hot acid treatment hydrolyzes ester bonds and releases fatty acids and other phosphoglyceride constituents.
- **Alkaline hydrolysis** : **Saponification :**

-Mild alkaline: Release of FAs in the form of soaps + skeleton (glycerol-P-alcohol(X) acid)

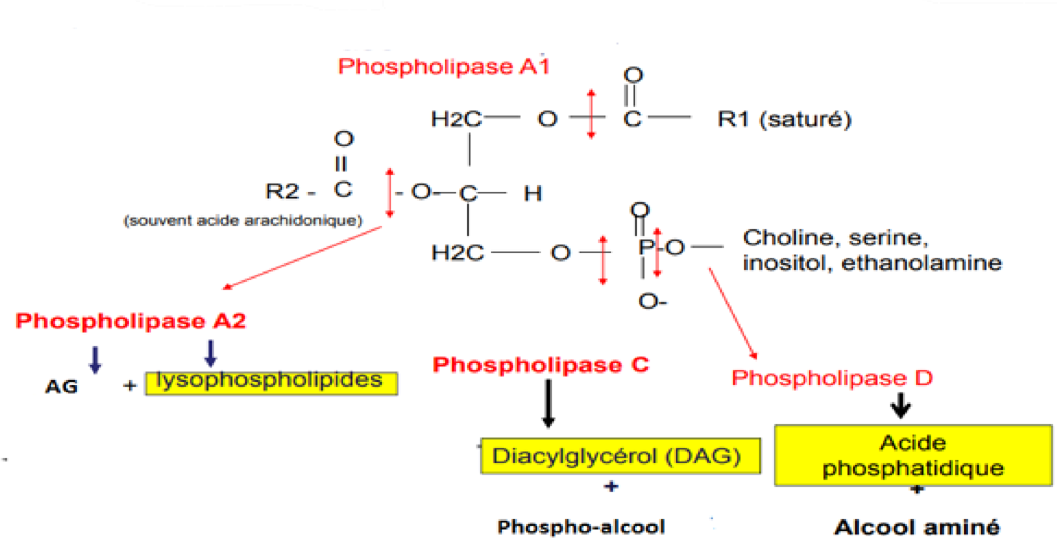
-Strong alkaline: Release of FAs in the form of soaps + Alcohol (X) + skeleton (glycerol-P acid).

2/Enzymatic hydrolysis

Enzymatic hydrolysis of phospholipids by **phospholipases** allows the release of biologically active compounds.

There are **four phospholipases, A1, A2, C, and D**, which hydrolyze ester bonds in a highly specific manner:

· If hydrolysed by **phospholipase A1**: saturated fatty acid + Lyso1 phospholipid
· If hydrolysed by **phospholipase A2**: unsaturated fatty acid + Lyso 2 phospholipid
· If hydrolysed **by phospholipase C:** diglyceride + phosphoryl-choline
· If hydrolysed **by phospholipase D**: phosphatidic acid + alcohol (X)

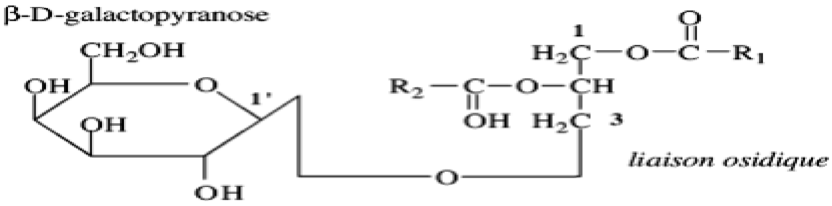


III/ GLYCEROGLYCOLIPIDS:

-As in glycerolipids, **C1** and **C2** of **glycerol** are **esterified** by fatty acids, but on **C3** an **oside** or **oligoside** is attached by its **hemiacetal carbon** via a **glycosidic bond**.

-These glyceroglycolipids are **abundant in plant membranes**.

Ex: 1,2-diacyl-[β-D-galactosyl-(1'-3)]-glycerol



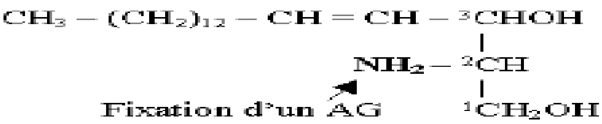
1,2-diacyl-[β-D-galactosyl-1'-3]-*sn*-glycérol

IV/ SPHINGOLIPIDS

A / Structure

-The **alcohol** is no longer glycerol but **Sphingosine**: a long-chain **amino alcohol** (C18):

- with **two alcohol** functions: one at **C1** and one at **C3**,
- and a **primary amine** function at **C2**.
- and a **double bond** at **C4 and C5**.

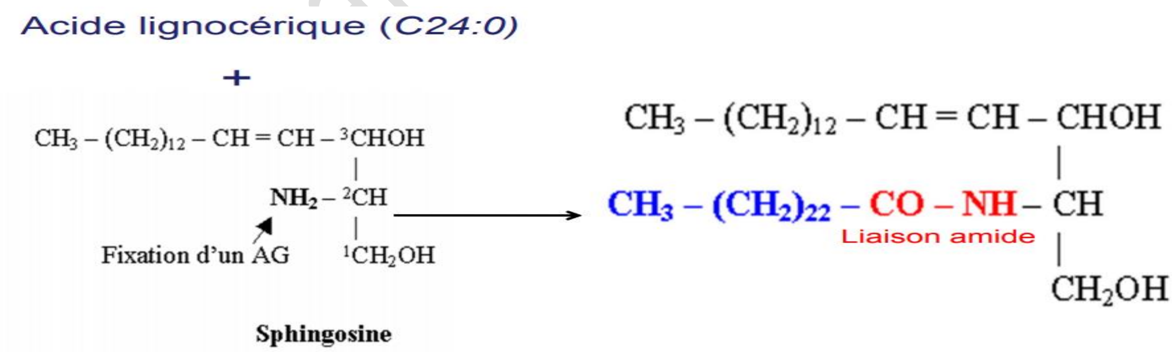


Sphingosine

-From sphingosine, we obtain derivatives:

1) Acylsphingosine or Ceramide

sphingosine + acide gras = céramide



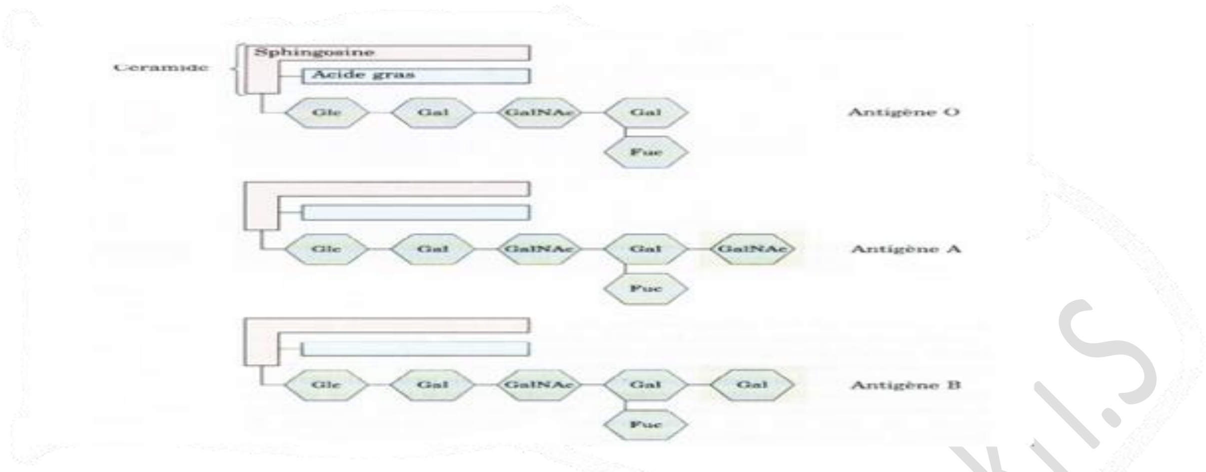
-The simplest sphingolipid is ceramide or acylsphingosine.

-The fatty acid is saturated and long-chain.

-Ceramide is an intracellular second messenger.

-**Blood groups** are **the combination** of **a carbohydrate** motif and **a ceramide**, which produces a glycolipid.

Blood groups



2) Sphingomyelins:

Céramide + phosphocholine = sphingomyéline

-They are composed of **sphingosine + fatty acid + phosphorylcholine.**

-The most common fatty acid is **lignoceric acid (C24:O).**

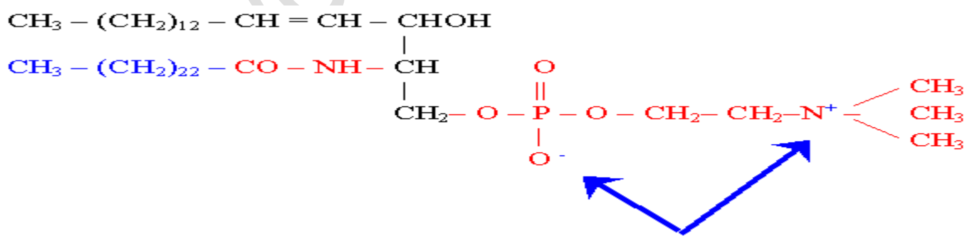
• At blood pH, the molecule **is ionized.**

• They are found in **nerve tissue** (myelin sheath) and in membranes.

Brain: **Gray matter:** C18:0 (stearic)

White matter: (C24:0; C24:1)

• Role in cell **transduction** and **neuronal activity**

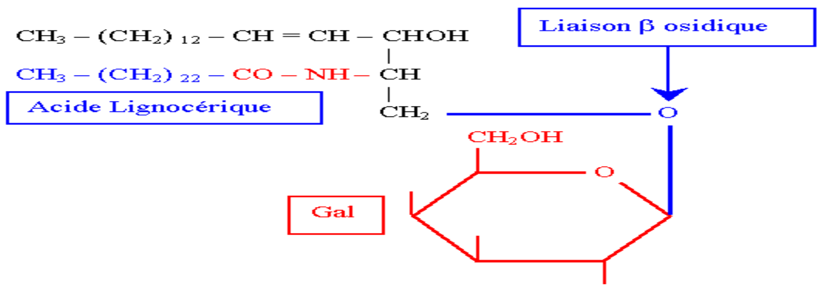


3) Glycolipids

a/Cerebrogalactosides or galactosylceramides

-They are composed of: **Sphingosine + FA + βD Galactose**

-Galactose is **linked** to the **primary alcohol of sphingosine** by a **β glycosidic bond.**



b/Cerebrosides or glucosylceramides

-They are composed of: sphingosine + fatty acid + βD glucose

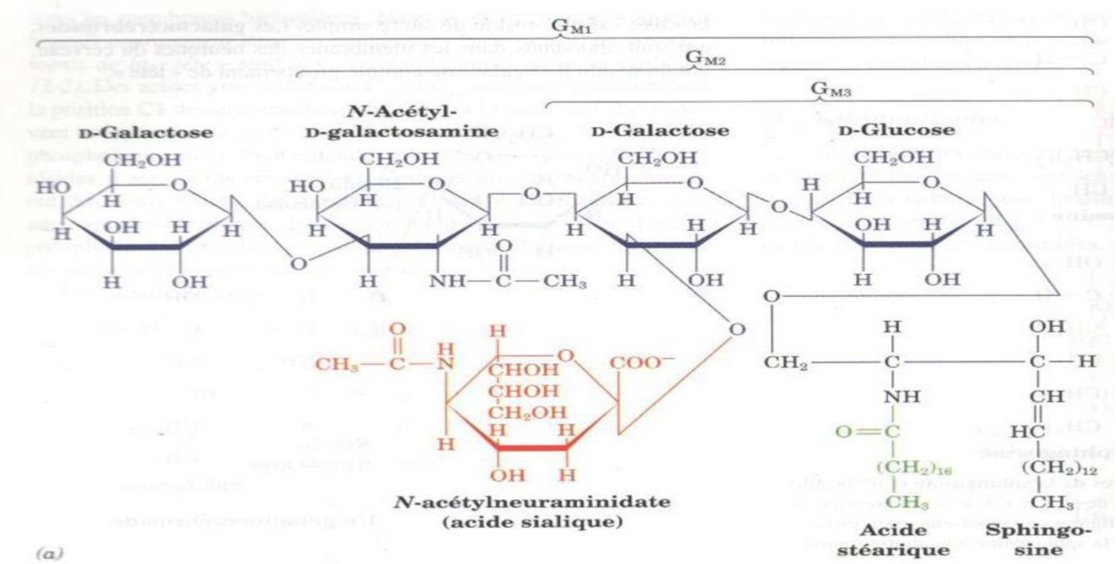
c/ Gangliosides or oligosylceramides

-They are composed of: **sphingosine + fatty acid + chain of several oses and oses derivatives (NANA) (= oligoside)**

-They are abundant in **ganglia**, hence **their name.**

-Gangliosides are acidic glycolipids whose oligosaccharide chain ends with **one or more sialic acid residues (NANA).**

-They are abundant in the **ganglion cells of the brain**, hence their name.



B/ Properties and roles of sphingolipids

- Structural role in membranes: sphingolipids are mainly constituents of cell membranes. They are inserted into the outer layer of the membrane, and the carbohydrate portion of glycosphingolipids is always on the outer surface of the cell (as with glycoproteins).
- Role in cell recognition:

  - an antigenic role: Ag of blood groups A, B, O.

- Role as intracellular mediators: sphingomyelin plays a role in the transduction of an extracellular signal inside the cell.